Lecture 9 (Thermo)

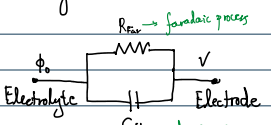
Tuesday, 7 February 2023 1:39 PM

Contributions of the Double Layer to Current

+ B-V describes current due to charge transfer rxns (faradaic)

+ Charge separation occurs in the double layer (charge accumulation) → polarizing an electrode
↓
manifesting as current (No flow of e^- so cannot be captured by B-V kinetics)

+ Qualitatively model using a circuit:

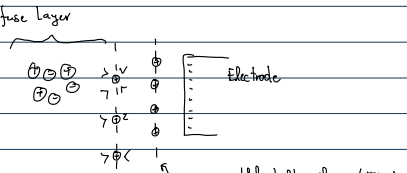


Capacitance [=] $F = \frac{C}{V}$;

$$i_{total} = i_{Far} + i_{dl} = \frac{V-\phi_e}{R_{Far}} + C_{dl}\frac{\partial(V-\phi_e)}{\partial t}$$; R_{Far} is given by $\frac{\partial \eta}{\partial i}$ (Ohm's law) ← B-V

@SS: $i_{dl} = 0$ → capacitive current = 0
↳ But DL still exist
} can impact transport of ions just by its presence

* How do we calculate C_{dl}?
↳ contains distribution of charge
↳ common model for a more accurate DL is the Helmholtz model



(weakly associated ions often solvated ∧ → H_2O)

(specifically adsorbed ions) considered "desolvated"

* Total charge must be balanced:

Gouy-Chapman Theory:

$$q_{electrode} = -(q_{IHP} + q_{OHP} + q_d)$$

We also know: $\frac{1}{C_{dl}} = \frac{1}{C_{IHP}} + \frac{1}{C_{OHP}} + \frac{1}{C_d}$,

where min $C_d = \frac{\varepsilon}{\lambda}$ ← permittivity; ↑ debye length

Thermodynamics

Voltage difference is an (electrochemical potential) ; What is electrochemical potential?
F
(That we measure across a load attached to EC)

Chemical Potential: $\mu_A = \left.\frac{\partial G}{\partial N_A}\right|_{T,P,N[j]}$; Relative values have meaning (i.e. to standard states)

work/energy associated w/ moving chemical species from one place to another.

$\mu_i = f(T, P, x_i)$

→ $W_{rev} = \mu_A(2) - \mu_A(1) = -\Delta G$

Electrical Potential:

Electrical Potential :

Energy associated w/ moving idealized point charges from one place to another (regardless of chemical state)

$$f = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2} \quad ; \quad W_{rev} = -\Delta G = F(V_2 - V_1) = \mu_e(2) - \mu_e(1)$$

A need for something more:

Issue: 1) Energy associated w/ moving e^- from one sphere to another: $\sim 10^8$ V [2cm Cu sphere, moving $1\times10^9 e^-$] (Case studies) but $\Delta\mu_e \approx 0$ (Adding small e^- to very large mass/mol of e^-)

2) Volta battery (uncharged electrodes to make electric current) → μ_e can't describe this since it ignores chemical state

but

μ can be used to understand the change.

∴ E-chem systems : need charged species in chemical state whose energies are affected by: + changes in P, T, composition + Electrical state → Electrochemical Potential!

$$\mu_j = \mu_j^c + z_j F \phi$$

μ_j^c: chemical potential; z_j: charge no.; ϕ: quasi-electrostatic potential

This is a formal description of echem potential

However, (constant of proportionality)

$$\mu_j = RT \ln(\gamma_j) \quad ; \quad \gamma_j = c_j f_j a_j$$

μ_j: Electrochemical activity; γ_j: activity; a_j: activity coefficient

← This definition utilizes difference in reference/standard states → More useful

→ This must depend on electrical state for ionic species

(But specific comb. of f_j are not dependent on electrical state) ←

$MX_2 \rightleftharpoons M^{2+} + 2X^-$

$$C_{MX_2} = C = \frac{C_{M^{2+}}}{1} = \frac{C_{X^-}}{2} \quad ; \quad \mu_{MX_2} = \mu_{M^{2+}} + 2\mu_{X^-}$$

$$\gamma_{MX_2} = \gamma_{M^{2+}} \gamma_{X^-}^2 = (C_{M^{2+}} f_{M^{2+}} a^\circ_{M^{2+}})(C_{X^-} f_{X^-} a_{X^-})^2$$

$$\gamma_{MX_2} = C^3 (a^\circ_{M^{2+}})(2a^\circ_{X^-})^2 (f_{M^{2+}} f_{X^-}^2)$$

All these species are not dependent on electrical state → So not dependent on electrical state! (Although individually they are)

Mean Activity Coeff

In general, $\bar{f}_\pm^{\nu} = f_+^{\nu_+} f_-^{\nu_-}$ for a general electrolyte

* Neutral comb. of ions will always have mean activity coeff independent of electric state

Electrolyte: $M_{\nu_+}X_{\nu_-} \rightleftharpoons \nu_+ M^{z+} + \nu_- X^{z-}$, $z_+\nu_+ + z_-\nu_- = 0$ (Electroneutrality)

Electroneutrality

of electric static

Why?

+ Physical props & rates → defined in terms of activities / not conc.

+ framework of echem sys